

Ian Tan

CURRICULUM VITAE

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Education:

- Doctor of Philosophy in Mathematics, Auburn University, 2025
Advisor: Dr. Luke Oeding
Thesis title: Orbits and Invariants in Quantum Information Theory
GPA: 4.0
- Bachelor of Science in Mathematics, Lee University, 2018
GPA: 3.98
Minor: Music

Research interests:

- Quantum Information Theory, Quantum Coding Theory, Entanglement

Publications/Preprints:

- I. Tan, *Transversal gates of the $((3,3,2))$ qutrit code and local symmetries of the absolutely maximally entangled state of four qutrits* (preprint), arXiv:2601.19677.
- I. Tan, *Classification of four-qubit pure codes and five-qubit absolutely maximally entangled states* (preprint), arXiv:2507.02185.
- H. Alpert, L. Barham, B. Freidin, I. Tan, A. Weiner, *Fast winning strategies in a generalized van der Waerden game*, *Integers* **26** (2026), arXiv:2309.11367.
- L. Oeding, I. Tan, *Four-qubit critical states*, *Journal of Physics A: Mathematical and Theoretical* **58** (2025), no. 26, arXiv:2410.08317.
- L. Oeding, I. Tan, *Tensor decompositions with applications to LU and SLOCC equivalence of multipartite pure states*, *SIAM Journal on Applied Algebra and Geometry* **9** (2025), no. 1, arXiv:2402.12542.
- I. Tan, *Counting rotational subsets of the circle under the angle-multiplying map*, *Rose-Hulman Undergraduate Mathematics Journal* **25** (2024), no. 2, arXiv:2207.03594.

Work Experience:

- Postdoc, Charles University, Prague Jan 2026-Present
- Teaching Assistant, Auburn University, Auburn, AL Spring 2021-Fall 2025
Calculus I and Calculus III

- Teacher, Greater Alabama Black Belt Region Louis Stokes Alliance for Minority Participation
Summers 2021-2024
- Teacher, EL Pathway Learning Centre, Malaysia
Mathematics and physics; various grade levels
2019-2020
- Teacher, Thomas MacLaren School, Colorado Springs, CO
10th grade Precalculus and 11th grade Calculus
2018-2019

Talks:

- Charles University, Condensed Matter Theory Seminar, *Lie algebras and quantum error correcting codes*, Feb 2026.
- University of Wisconsin-Madison, 2025 SIAM Conference on Applied Algebraic Geometry Minisymposium: Quantum Information Theory and Geometry, *Vinberg theory of the four-qubit state space and applications*, July 2025
- University of Tennessee Knoxville, SIAM-SEAS 2025 Minisymposium: Applied Algebra and Geometry, *Tensor decompositions with applications to LU and SLOCC equivalence of multipartite pure states*, March 2025
- Georgia Institute of Technology, Algebra Seminar, *Tensor decompositions with applications to LU and SLOCC equivalence of multipartite pure states*, Jan 2025.
- University of California San Diego, Optimization and Data Science Seminar, *Tensor decompositions with applications to LU and SLOCC equivalence of multipartite pure states*, Nov 2024.
- University of Alabama at Birmingham, SIAM Student Chapter Seminar, *Entanglement and the Mermin-Peres magic square game*, Nov 2024.
- University of Idaho, Algebra Seminar, *Tensor decompositions with applications to LU and SLOCC equivalence of multipartite pure states*, Oct 2024.
- Auburn University, Algebra Seminar, *Four-qubit critical states*, Oct 2024.
- University of Georgia, Algebra Seminar, *Tensor decompositions with applications to LU and SLOCC equivalence of multipartite pure states*, Sept 2024.
- Auburn University, Invariant @60 Conference, *Tensor decompositions with applications to LU and SLOCC equivalence of multipartite pure states*, May 2024.
- Auburn University, Algebra Seminar, *Tensor factorizations and orbit classification*, Mar 2024.
- Louisiana State University, Fourth Biennial LSU Music Colloquium, *Linear Algebraic Applications to Twelve-Tone Tonality*, May 2017

Graduate awards:

- Spring 2025 Don and Sandy Logan Endowed Fellowship
- 2023 Dr. J. Earl Perry Endowed Summer Graduate Award
- 2022 Dr. J. Earl Perry Endowed Summer Graduate Award
- 2020 Calder Endowed Graduate Award

Outreach:

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| • Mentor, Directed Reading Program | Spring 2025 |
| • Topological Data Analysis Seminar | Fall 2024 |
| Co-organizer with Mauricio Montes | |
| • Mentor, Directed Reading Program | Spring 2024 |
| • Mentor, Directed Reading Program | Spring 2023 |
| • Volunteer, Alabama Science State Olympiad | Spring 2023 |
| • Volunteer, Alabama Science State Olympiad | Spring 2022 |